Table of Contents

1. PURPOSE 2

2. SCOPE 2

3. REFERENCES 3

4. DEFINITIONS 3

5. RESPONSIBILITIES 3

6. WHEN METHOD VALIDATION IS REQUIRED 4

7. VALIDATION PLANNING 4

8. PERFORMANCE CHARACTERISTICS TO BE EVALUATED 5

9. EXPERIMENTAL DESIGN 5

10. STATISTICAL EVALUATION 6

11. ACCEPTANCE CRITERIA 6

12. MEASUREMENT UNCERTAINTY 6

13. VALIDATION REPORT 7

14. AUTHORIZATION AND IMPLEMENTATION 7

15. RECORDS AND DOCUMENT CONTROL 7

16. REVIEW AND REVALIDATION 7

17. REVISION TABLE 8

CONTROLLED DOC

1. PURPOSE

The purpose of this procedure is to define the systematic approach used by the laboratory to validate test methods to ensure that they are fit for their intended use, technically sound, and capable of producing reliable, accurate, and traceable results, in compliance with:

- ISO/IEC 17025:2017 (Clauses 7.2, 7.6, 7.7)
- SANAS TR 26 (Method Validation and Verification Guidelines)
- Applicable international and national test standards (ISO, ASTM, SANS)

2. SCOPE

This procedure applies to:

- Non-standard methods
- Laboratory-developed methods
- Standard methods used outside their intended scope
- Modified standard methods
- Verification of standard methods, where required by SANAS TR 26

Typical coal laboratory parameters include, but are not limited to:

- Ash content
- Moisture
- Volatile matter
- Calorific value
- Total sulphur
- Float and sink analysis

CONTROLLED DOC

3. REFERENCES

- ISO/IEC 17025:2017, Clause 7.2 – Selection, Verification and Validation of Methods
- ISO/IEC 17025:2017, Clause 7.6 – Evaluation of Measurement Uncertainty
- ISO/IEC 17025:2017, Clause 7.7 – Ensuring the Validity of Results
- SANAS TR 26 – Method Validation and Verification
- ISO 13528 – Statistical methods for PT schemes
- ISO 17034 – Reference material production (where CRMs are used)

4. DEFINITIONS

Validation: Confirmation through objective evidence that requirements for a specific intended use are fulfilled.

Verification: Confirmation that a standard method performs as expected under laboratory conditions.

Performance Characteristics: Parameters evaluated to demonstrate method suitability (e.g., precision, accuracy, bias).

5. RESPONSIBILITIES

Role	Responsibility
Laboratory Manager / Supervisor	Approval of validation plan, review of results, and authorization
Laboratory Analysts	Execution of validation experiments
Operations Manager	Compliance review, document control, record retention

CONTROLLED DOC

6. WHEN METHOD VALIDATION IS REQUIRED

Method validation shall be conducted when:

- A new method is introduced
- A standard method is modified
- A method is used outside its published scope
- Equipment, reagents, or critical parameters change
- Results from QC, PT, or audits indicate potential method issues
- Confirmation that a standard method performs as expected under laboratory conditions.

(Standard methods used strictly as published may require verification rather than full validation, in accordance with SANAS TR 26.)

7. VALIDATION PLANNING

7.1 Validation Plan Approval

Before validation, a Method Validation Plan shall be prepared and approved, defining:

- Method name and reference
- Measurand and reporting units
- Intended use and sample matrix
- Performance characteristics to be evaluated
- Acceptance criteria
- Experimental design
- Statistical tools to be used

CONTROLLED DOC

8. PERFORMANCE CHARACTERISTICS TO BE EVALUATED

The following characteristics shall be assessed, as applicable to the method and measurand:

8.1. Accuracy

Evaluated using:

- Certified Reference Materials (CRMs), where available
- Comparison to reference or consensus values
- Interlaboratory comparison data (PT results)

8.2. Precision

Assessed as:

- **Repeatability** (same analyst, same equipment, short interval)
- **Intermediate precision** (different days, analysts, or instruments)

Reported as standard deviation and %RSD.

8.5. Linearity and Working Range (where applicable)

Assessment across the expected concentration range to confirm proportional response.

8.6. Detection and Quantification Limits (if relevant)

Established using statistical or empirical approaches where required.

8.7. Robustness

Assessment of the effect of small, deliberate variations in method parameters.

9. EXPERIMENTAL DESIGN

- A representative number of samples shall be analyzed.

CONTROLLED DOC

- Replicate measurements shall be performed according to the method's precision requirements.
- Measurements shall be performed using **validated equipment**, calibrated and traceable in accordance with ISO/IEC 17025 Clause 6.5.
- Environmental conditions shall be monitored and controlled where relevant.

10. STATISTICAL EVALUATION

- Appropriate statistical tools shall be applied (mean, standard deviation, %RSD, ANOVA, bias assessment).
- Outlier testing (e.g., Grubbs' test) shall only be applied where justified and documented.
- Robust statistics may be used where data behavior warrants (SANAS TR 26).

11. ACCEPTANCE CRITERIA

Acceptance criteria shall be defined **prior to validation** and may include:

- Precision limits derived from method standards or historical performance
- Bias within stated uncertainty or method tolerance
- Comparison against published method performance or PT statistics

Failure to meet acceptance criteria shall trigger investigation, corrective action, or method revision.

12. MEASUREMENT UNCERTAINTY

Measurement uncertainty shall be:

- Evaluated in accordance with ISO/IEC 17025 Clause 7.6
- Based on validation data, QC performance, CRMs, PT data, and method variability
- Documented and reviewed as part of the validation outcome

CONTROLLED DOC

13. VALIDATION REPORT

A formal **Method Validation Report** shall be issued, including:

- Summary of experiments performed
- Raw and processed data
- Statistical evaluation
- Comparison against acceptance criteria
- Statement of method fitness for intended use
- Authorization by Laboratory Manager / Supervisor

14. AUTHORIZATION AND IMPLEMENTATION

Only after successful validation and formal approval may the method be:

- Released for routine testing
- Included in the laboratory's scope of accreditation (where applicable)
- Used for reporting customer results

15. RECORDS AND DOCUMENT CONTROL

All validation records shall be:

- Controlled in accordance with the laboratory's document control procedure
- Retained for the period defined by SANAS and internal policies
- Made available during SANAS assessments

16. REVIEW AND REVALIDATION

Methods shall be reviewed periodically or when:

- QC trends indicate instability
- PT performance deteriorates

CONTROLLED DOC

- Method conditions change significantly

17. REVISION TABLE

Revision no	Content changed	Reason for revision	Date changed

CONTROLLED DOC